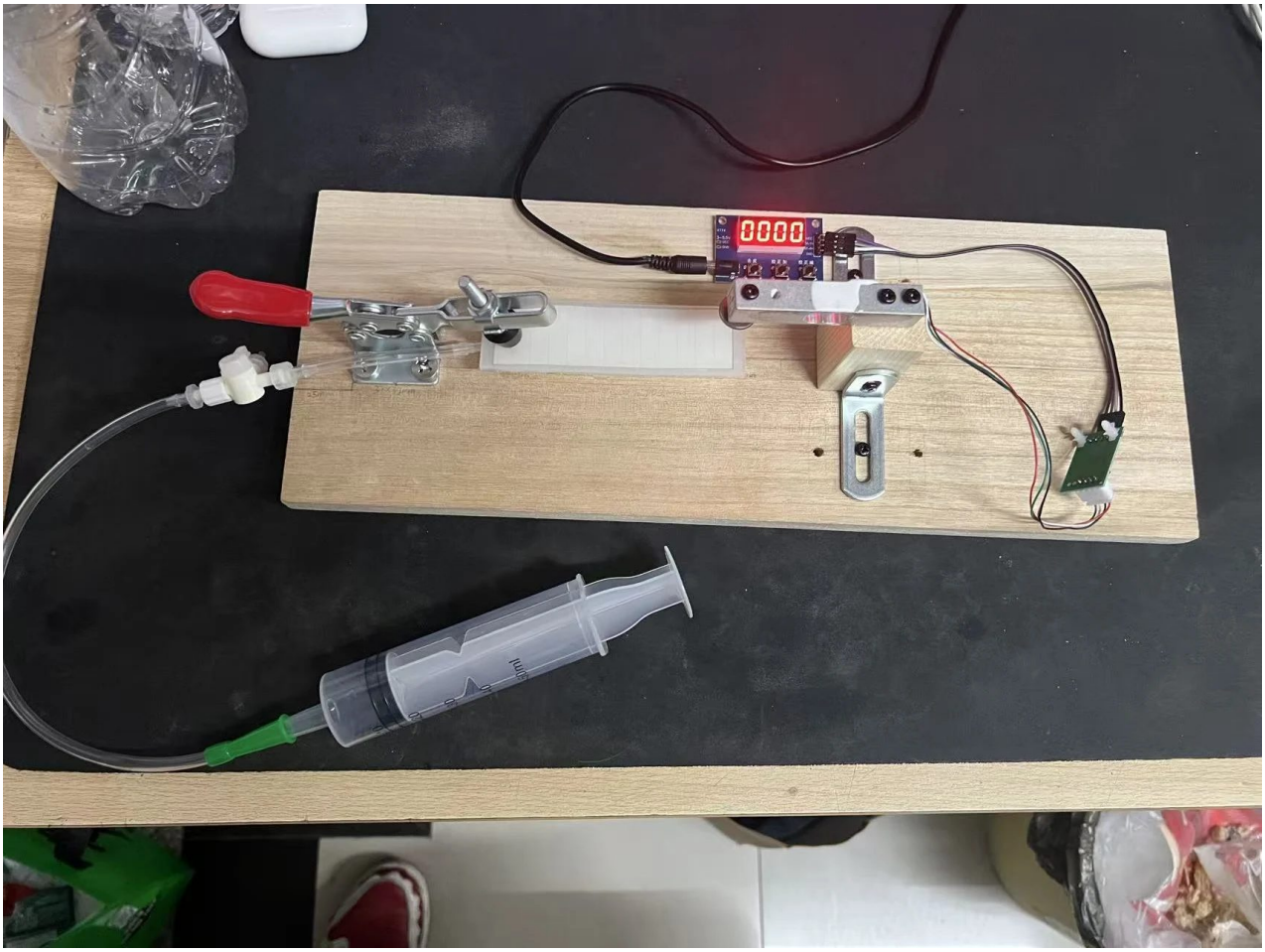
Test Rig for Soft Fluidic Actuators

Yuzhou Li | Undergraduate dissertation project | 2025–2026

Project objective

Develop a low-cost, adjustable test platform to measure peak blocked force from different soft fluidic actuator samples using accessible laboratory components.



Final platform: wooden base, adjustable clamp and sensor mounting, manual syringe inflation, and an electronic scale module. Photograph supplied by Yuzhou Li.

Engineering work

Designed and assembled the mechanical fixture and pneumatic arrangement. Investigated leakage in the original pressure-gauge layout and simplified the connections. Replaced an unstable load-cell, INA125P and Arduino sensing arrangement with a commercial electronic scale module.

Measurement method

After fixing and aligning the actuator, zero the scale, inject a selected air volume and record the displayed peak mass on video. Convert grams to force using $F = (m / 1000) \times 9.81$. A known 60 g mass was used for a basic scale check.

Measurement Results

Rounded values reproduced from Tables 4–5 of Yuzhou Li's dissertation.

Sample	Air (mL)	Mean force (N)	SD (N)	Repeats
PneuNet	15	0.164	0.020	3
PneuNet	20	0.376	0.020	3
PneuNet	25	0.644	0.015	3
PneuNet	30	0.870	0.046	3
PneuNet	35	0.991	0.029	3
PneuNet	40	0.975	0.154	5
Fibre-reinforced	35	0.144	0.020	3
Fibre-reinforced	40	0.294	0.035	3
Fibre-reinforced	45	0.428	0.015	3
Fibre-reinforced	50	0.549	0.035	3
Fibre-reinforced	55	0.795	0.026	3
Fibre-reinforced	60	0.952	0.125	5

Interpretation

The platform recorded useful force–volume trends for both actuator structures. The highest mean peak blocked force was 0.991 N for the PneuNet sample at 35 mL. Variation increased at the upper test volumes, where manual inflation and contact alignment could affect the readings.

Scope and limitations

Internal pressure was not measured; these are force–volume results, not pressure-controlled characterisation. The two samples differ in geometry and inflation range, so the data demonstrate the test rig rather than provide a controlled ranking of actuator designs. The 60 g check was not a full calibration.

Source and acknowledgements

Adapted from Yuzhou Li, Test rig for soft fluidic actuators (2026), supervised by Dr Nicolas Herzig, Sussex Artificial Intelligence Institute, Zhejiang Gongshang University. Actuator fabrication methods were informed by the Soft Robotics Toolkit PneuNets and fibre-reinforced actuator guides. No external literature images are reproduced.